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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 02/18/2025 |
| **PI Name:** | Marquez, Jessica J. Ph.D. | | |
| **Project Title:** | HCAAM VNSCOR: Crew Autonomy through Self-Scheduling: Guidelines for Crew Scheduling Performance Envelope and Mitigation Strategies | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | NASA CENTER | **Phone:** | 650-604-6364 |
| **Organization Name:** | NASA Ames Research Center | | |
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| **City:** | Moffett Field | **State:** | CA |
| **Zip Code:** | 94035 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C |
| **Start Date:** | 04/15/2019 | **End Date:** | 09/30/2025 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 09/30/2025 per A. Beitman/JSC (Ed., 2/21/23) | | |
| **Key Personnel Changes/Previous PI:** | February 2020 report: Mr. Steven Hillenius (Co-Investigator) left NASA. Dr. Tamsyn Edwards is replacing Mr. Hillenius as Co-I. Dr. Edwards works at NASA Ames as part of San Jose University Research Foundation. February 2021 report: Dr. John Karasinski is now a Co-I. February 2022 report: Dr. Edwards has left NASA. March 2023 report: Dr. Karasinski has changed affiliations. February 2024 report: Dr. Karasinski has changed affiliations. | | |
| **COI Name (Institution):** | Bresina, John Ph.D. ( NASA Ames Research Center )<br>Gregory, Kevin M.S. ( San Jose State University Research Foundation )<br>Zheng, Jimin M.S. ( San Jose State University Research Foundation )<br>Edwards, Tamsyn Ph.D. ( San Jose State University Research Foundation )<br>Karasinski, John Ph.D. ( NASA Ames Research Center ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |

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| **Performance Goal Text:** | |
| **Task Description:** | This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR).<br>As NASA considers long-duration exploration missions (LDEMs), it is envisioned that crew will behave more autonomously as compared to low-Earth orbit missions. In this space environment, crew will have better and more timely insight as to how best to manage their own schedule, minimizing idle time as they wait for Mission Control Center (MCC) to respond or react to a delay in activity execution. Moreover, crew must be able to self-schedule, i.e., reschedule their own timeline without creating violations. NASA currently has not characterized crew performance for self-scheduling; specifically, non-expert human performance for the task of planning and scheduling has not been characterized experimentally. The focus of this proposal is to quantify crew performance envelope for the task of planning and scheduling as a function of plan complexity, and develop mitigations that are aimed at improving performance in the face of complex planning requirements. With regards to crew performance, we will study the relationship between planning efficiency, effectiveness, crew situation awareness, trust in planning software, and plan complexity. Once a performance envelope has been identified, we will shift our research emphasis to develop and evaluate countermeasures that mitigate adverse effects on performance. These mitigations will be evaluated in analogs and recommended countermeasures will be put forward if crew performance improves as compared to the baseline. Finally, based on research results, we will recommend corresponding standards and guidelines appropriate for autonomous crew in LDEMs. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | NASA currently has not characterized crew performance for self-scheduling; specifically, novice human performance for the task of planning and scheduling has not been characterized experimentally. As a result of this research, we will quantify the user performance envelope for the task of planning and scheduling, which impacts many jobs both on Earth and in spaceflight. The knowledge gained from our research can be generalized to benefit our understanding on how to improve roles that require planning and scheduling, such as project planning, personnel scheduling, and operational management. Our research will also contribute to developing the next generation of planning, scheduling, and execution software tools for NASA. |
| **Task Progress:** | Task Progress Summary Year 5<br>As NASA considers long-duration exploration missions (LDEMs), it is envisioned that crew will behave more autonomously as compared to low-Earth orbit missions. In this space environment, crew will have better and more timely insight how to best manage their own schedule, minimizing idle time as they wait for Mission Control Center (MCC) to respond or react to a delay in activity execution. Moreover, crew must be able to self-schedule—that is, reschedule their own timeline without creating violations. NASA currently has not characterized crew performance for self-scheduling; specifically, non-expert human performance for the task of planning and scheduling has not been characterized experimentally. The focus of this research is to quantify crew performance envelope for the task of planning and scheduling as a function of plan complexity, and develop mitigations aimed at improving performance in the face of complex planning requirements. With regards to crew performance, we will study the relationship between planning efficiency, effectiveness, crew situation awareness, trust in planning software, and scheduling task complexity. Once a performance has been characterized, we will shift our research emphasis to develop and evaluate countermeasures that mitigate adverse effects on performance. These mitigations will be evaluated in analogs and recommended countermeasures will be put forward if crew performance improves as compared to the baseline. Finally, based on research results, we will recommend corresponding standards and guidelines appropriate for autonomous crew in LDEMs.<br><br>For Year 6 (4/2024 – 4/2025), our research team completed data collection for NASA Human Exploration Research Analog (HERA) Campaign 7. We have started post-processing and analyzing this data for our dependent variables (self-scheduling efficiency, effectiveness, usability, workload, trust, and collaboration). We have also collected data for a controlled-lab experiment to assess self-scheduling performance while using the countermeasure aids available in HERA. Results will be available in Year 7. We conducted an exploratory analysis of HERA C6 data spontaneous self-scheduling, indicating that 1 out of 4 flexible activities were rescheduled during HERA C6. Differences between role assignment and missions are present. Finally, a preliminary set of self-scheduling standards and guidelines has been drafted. Within Year 6, ongoing results were published in six conferences. |
| **Bibliography Type:** | Description: (Last Updated: 05/06/2026) |
| **Abstracts for Journals and Proceedings** | Marquez JJ. "What is the role of usability and trust in autonomy?" 76th Department of Defense Human Factors Engineering Technical Advisory Group, Marshall Space Flight Center, Huntsville, Alabama, April 22-25, 2025.<br>Abstracts. 76th Department of Defense Human Factors Engineering Technical Advisory Group, Marshall Space Flight Center, Huntsville, Alabama, April 22-25, 2025. , Apr-2024 |
| **Abstracts for Journals and Proceedings** | Marquez JJ, Karasinski JA, Shelat S, Abbott R, Zheng J, Bresina J. "Crew autonomy through self-scheduling: Guidelines for crew scheduling performance envelope and mitigation strategies." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025.<br>2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |
| **Articles in Peer-reviewed Journals** | Shelat S, Marquez JJ, Zheng J, Karasinski JA. "Collaborative system usability in spaceflight analog environments through remote observations." Appl Sci. 2024 Feb 28;14(5):2005. https://doi.org/10.3390/app14052005 , Feb-2024 |
| **Papers from Meeting Proceedings** | Vanderdonckt J, Vatavu RD, Manon J, Saint-Guillain M, Lefevre P, Marquez JJ. "Might as Well Be on Mars: Insights on the Extraterrestrial Applicability of Interaction Design Frameworks from Earth." 2024 CHI Conference on Human Factors in Computing Systems (CHI EA), Honolulu, HI, May 11-16, 2024.<br>2024 CHI Conference on Human Factors in Computing Systems (CHI EA), Honolulu, HI, May 11-16, 2024. , May-2024 |

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| **Papers from Meeting Proceedings** | Karasinski JA, Shelat S, Marquez JJ. "Validation of Self-Scheduling Countermeasures in NASA's HERA Campaign 6." 2025 AIAA SciTech Forum, Orlando, FL, January 6-10, 2025.<br>2025 AIAA SciTech Forum, Orlando, FL, January 6-10, 2025. , Jan-2025 |